

Roots of polynomial functions



1 Determine whether the following values are zeros of the function $f(x) = x^4 - 7x^3 + 12x^2 + 4x - 16$:

a $x = 5$

b $x = 4$

c $x = -1$

d $x = 1$

2 For each of the following functions:

i State the zeros of the function.

ii State the multiplicity of each zero.

a $f(x) = (x + 6)^3 (x - 3) (x + 9)$

b $f(x) = (x + 1)^2 (x - 6) (x + 2)^5$

c $f(x) = -3(x - 6)^4 (x + 2)^5 x^3$

d $f(x) = x^3 - 3x^2 - 5x + 15$

3 Determine whether each of the following equations have a root of multiplicity 4:

a $(x + 8)^4 (x + 3) = 0$

b $(x + 8) (x + 3)^3 = 0$

c $(x + 8) (x + 3) = 0$

d $(x + 8)^3 (x + 3) = 0$

4 Determine whether each of the following equations have a root of multiplicity 3:

a $(x + 1) (x^2 + 2x + 1) (x + 3) = 0$

b $(x + 1)^3 (x + 3) = 0$

c $(x + 1)^2 (x + 1) (x + 3) = 0$

d $(x + 1) (x - 9) (x + 3) = 0$

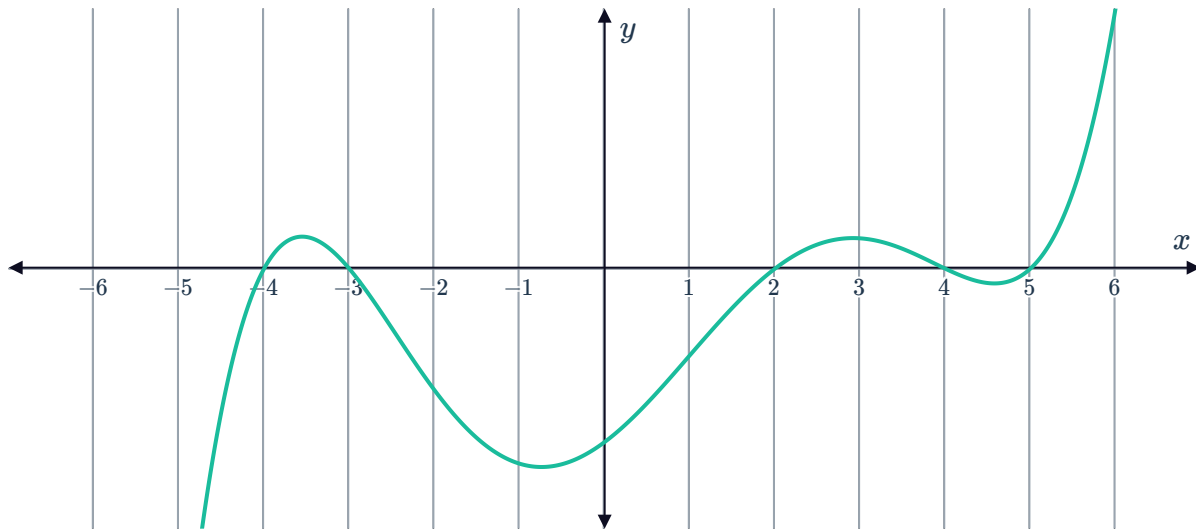
e $(x + 2) (x - 8)^3 (x - 1) = 0$

f $(x^3 + 1) (x - 2) (x + 1) = 0$

g $(x + 3) (x - 1)^2 (x - 3) = 0$

h $(x^2 - 2x + 1) (x - 1) (x + 9)^3 = 0$

5 Consider the following graph of a function

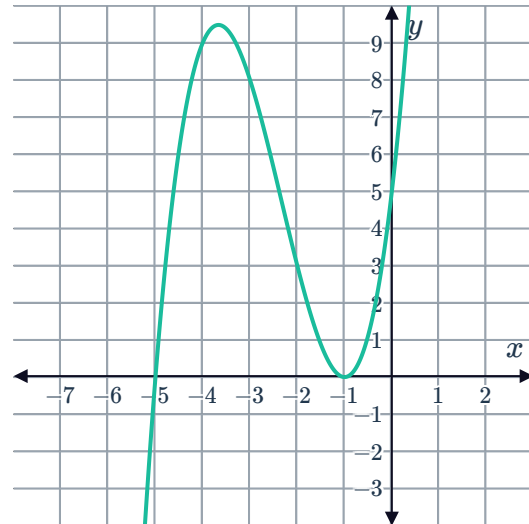


- a State all the linear factors of $f(x)$.
- b Write an expression for a polynomial $h(x)$ with a leading coefficient of 1, as a product of linear factors that has exactly the same roots as $f(x)$.
- c Determine whether the following has roots which include all the roots of $f(x)$:
 - i $2(x+4)(x-5)(x+3)(x-4)(x-2)$
 - ii $(x+4)(x+3)(x-2)^2(x-5)$
 - iii $4(x-4)(x-3)(x+2)(x+4)(x+5)$
 - iv $(x+4)(x+3)(x-2)(x-4)(x-5)(x+7)(x-11)$
- d Apart from $f(x)$, how many other polynomials are there with the same roots and the same graph as $f(x)$?

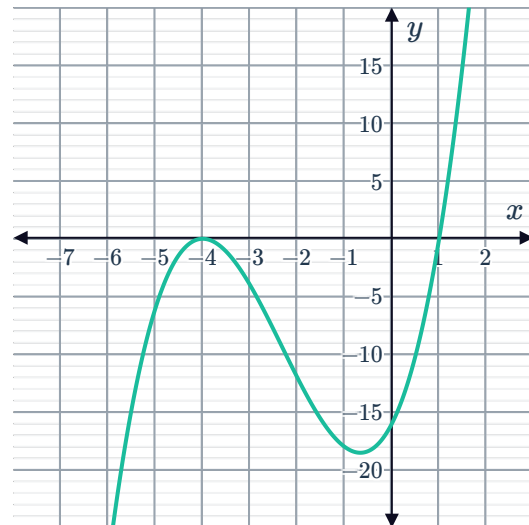


Equations of polynomial functions

- 6 Consider the following graph of a cubic function:
- a State the x -intercepts of the function.
 - b State the x -intercept that has a double root.
 - c State the equation of the cubic function.



- 7 Find the equation of the graphed cubic function:



8 Find the equation of the following cubic functions in standard form:



a

- Has real coefficients
- Zeros at $x = 5$, $x = -3$, $x = 1$
- When $x = -2$, $y = -84$.

b

- Has real coefficients
- Zeros at $x = 0$, $x = 2$, $x = 5$
- When $x = -1$, $y = 72$.

c

- Has real coefficients
- Zeros at $x = 0$, $x = 6$, $x = -6$
- When $x = -4$, $y = 20$.

d

- Has real coefficients
- Zeros at $x = -5$, $x = 4$, $x = -1$
- When $x = 2$, $y = -126$.

e

- Has real coefficients
- Zeros at $x = 4$, $x = 1$, $x = 5$
- When $x = 2$, $y = 18$.

f

- Has real coefficients
- Zeros at $x = -1$, $x = -4$, $x = 3$
- When $x = -5$, $y = 64$.

g

- Has real coefficients
- Zeros at $x = -3$, $x = -1$, $x = -4$
- When $x = -2$, $y = -10$.

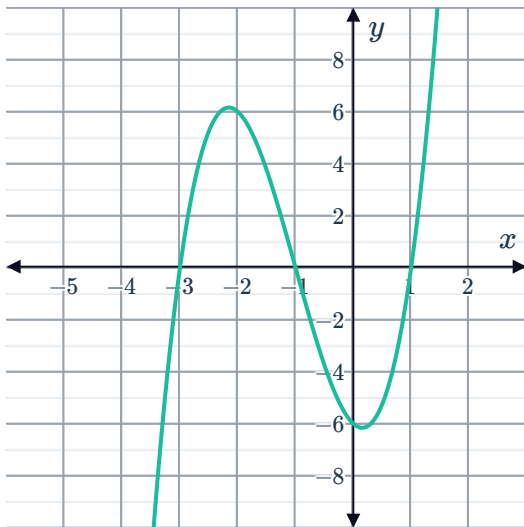
h

- Has real coefficients
- Zeros at $x = 4$, $x = 1$, $x = -2$
- When $x = 3$, $y = 50$.

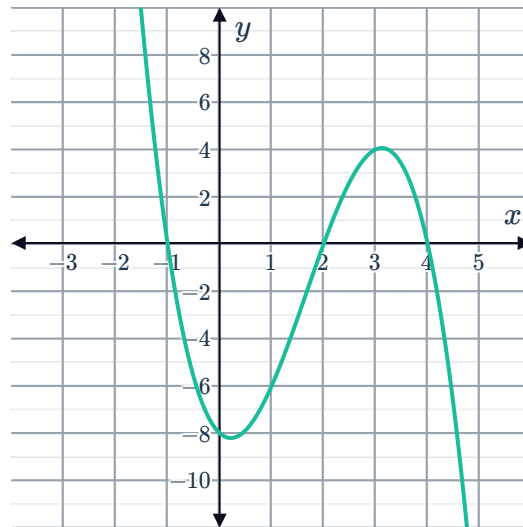
9 Evan graphed a cubic function, $f(x) = ax^3 + bx^2 + cx + d$, and determined the roots of $f(x)$ to be ± 4 and 5. Find the value of b if $a = 1$.

10 Find the equation of the following graphed functions in factored form:

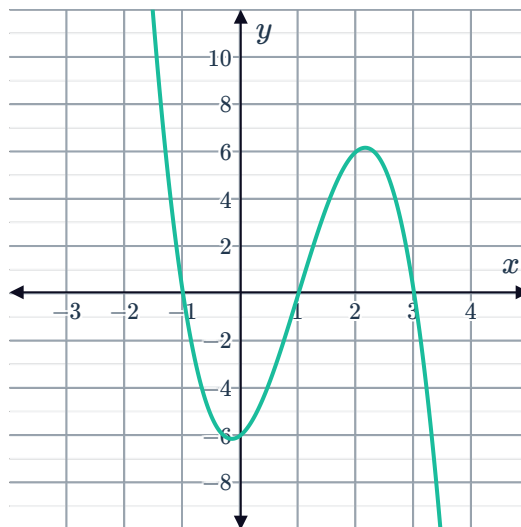
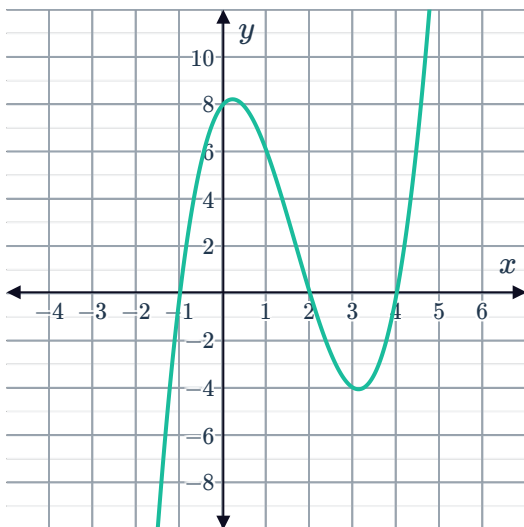
a



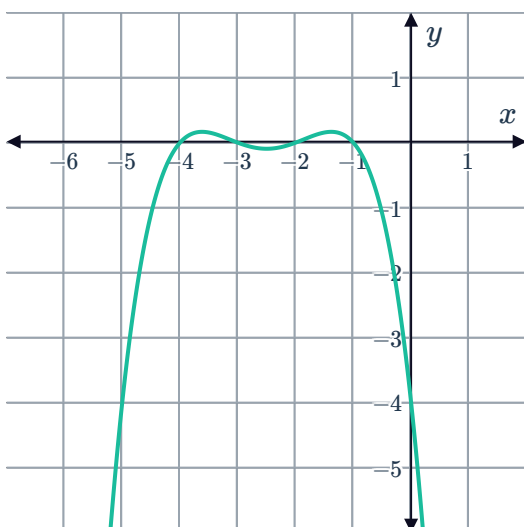
b



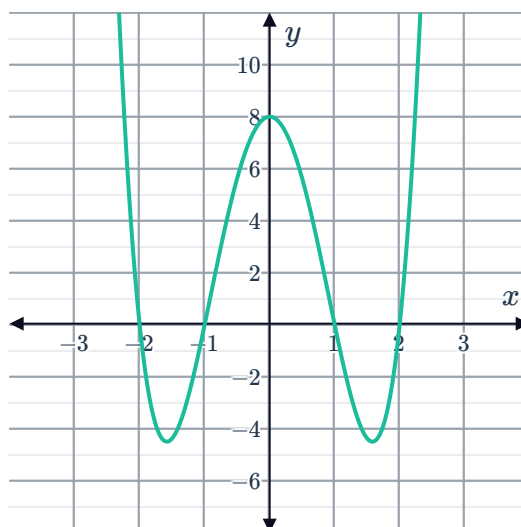
c



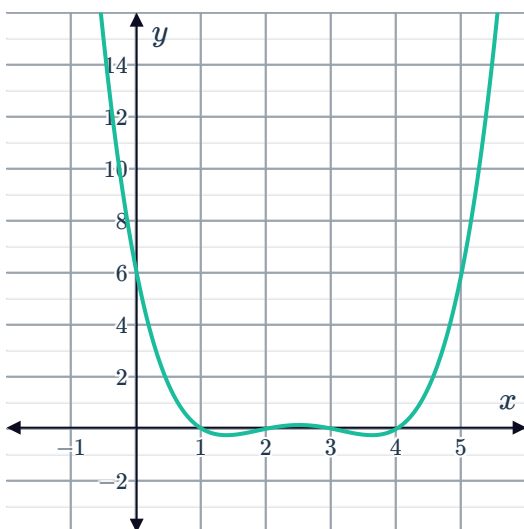
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